

TransMamba-Long: A Parameter-Efficient Hybrid Architecture for Long-Sequence Text Classification

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Abstract

This study investigated the efficiency of a hybrid Transformer-State Space Model for long-sequence text classification to overcome the quadratic computational bottleneck of traditional self-attention mechanisms. The investigation utilized a pretrained Longformer encoder fused with a 4-layer Mamba decoder via multi-head cross-attention and optimized using Low-Rank Adaptation (LoRA). The models were evaluated on the IMDB dataset with a maximum context length of 1024 tokens and compared against standard, native, and overclocked Transformer baselines. The 4-layer TransMamba-Long achieved a classification accuracy of 94.33% with a training time of approximately 18.5 minutes per epoch. In contrast, the native BART-base and overclocked DistilBERT required nearly 50 minutes, yielding lower accuracies of 89.07% and 91.36%, respectively. The results established that the hybrid architecture successfully bypassed the $\mathcal{O}(n^2)$ complexity bottleneck while maintaining state-of-the-art accuracy. We concluded that the 4-layer Mamba configuration provides an optimal trade-off between parameter efficiency and performance for long-document analysis, although its benefits remain task-dependent and are less pronounced on short-text sequences.

Keywords: Long-Sequence Classification, Hybrid Architecture, State Space Models, Mamba, Parameter-Efficient Fine-Tuning, Low-Rank Adaptation.

1 Introduction

Sentiment analysis and document classification on long-form text data remain cornerstone challenges in contemporary Natural Language Processing (NLP). Prior to this research, it was well-established that Transformer-based models exhibit a quadratic complexity bottleneck ($\mathcal{O}(n^2)$) with respect to the input sequence length. This constraint stems directly from the global self-attention mechanism, which requires every token within a sequence to compute an alignment score with every other token. Consequently, standard baseline models, such as BERT and DistilBERT, typically enforce a rigid maximum sequence limit of 512 tokens. When forced to process text fields that extend beyond this boundary, these models resort to aggressive truncation, which results in severe information loss, particularly when critical semantic qualifiers or concluding sentiments are embedded deep within long texts.

To resolve this issue without incurring astronomical hardware requirements or requiring enterprise-level compute clusters, recent advancements have explored State Space Models (SSMs). Among these architectures, Mamba has emerged as a highly promising alternative due to its selection mechanism and hardware-aware implementation, scaling linearly ($\mathcal{O}(n)$) with context length. However, the specific conditions under which Mamba improves discriminative text classification over strong Transformer baselines have remained insufficiently understood. Most existing hybrid Transformer-SSM architectures have been primarily optimized and evaluated on autoregressive generative or reasoning benchmarks, leaving their application to downstream sequence classification tasks largely unexplored.

This study aimed to address this critical gap by conducting a rigorous empirical investigation of a hybrid architecture that combines the localized representation capability of a Transformer encoder with the highly efficient sequential processing of a Mamba decoder. The primary approach involved evaluating different feature fusion strategies and model depths across extended sequence horizons. We hypothesized that a parameter-efficient hybrid framework could maintain state-of-the-art accuracy on long documents while significantly reducing computational latency and hardware footprints.

The principal findings revealed that our proposed 4-layer TransMamba-Long configuration achieved superior classification accuracy and drastically lower training latency on a 1024-token benchmark compared to traditional full self-attention baselines. This paper presents a comprehensive breakdown of the trade-offs involved, demonstrating that deep structural efficiency can conquer long-document analysis without the need for massive parameter duplication or exhaustive computational resources.

2 Materials

The empirical evaluation conducted in this study relied upon a combination of benchmark datasets, specialized neural architectural components, and controlled hardware configurations.

2.1 Datasets and Data Processing

The primary long-sequence data source utilized was the **IMDB Large Movie Review Dataset**, a standard benchmark for binary sentiment classification. This dataset consists of 50,000 highly polar movie reviews, split equally into 25,000 samples for model training and 25,000 samples for testing and evaluation. The dataset is characterized by a significant proportion of long-form text records, making it highly suitable for testing models under extended sequence limits where traditional Transformers hit computational boundaries.

To provide a robust contrast against short-text tasks and analyze the task-dependent nature of the hybrid architecture, subsets of the **GLUE (General Language Understanding Evaluation) Benchmark** were also integrated during preliminary testing stages, specifically the Stanford Sentiment Treebank (SST-2) and the Recognizing Textual Entailment (RTE) datasets.

2.2 Architectural Components and Sub-networks

The hybrid framework was constructed using two primary structural core modules:

- **Encoder Component:** Pretrained weights were sourced from the *allenai/longformer-base-4096* checkpoint via the Hugging Face Transformers library. This base model was selected because its localized sliding-window attention mechanism naturally reads extended sequences without suffering from early memory explosion.
- **Decoder Component:** The linear sequence tracking layers were initialized using the official *mamba-ssm* library (version 1.2.0), utilizing specialized selective scan operators.
- **Parameter-Efficient Module:** Low-Rank Adaptation (LoRA) modules were obtained from the *peft* library to restrict the trainable parameter boundaries.

2.3 Hardware and Computing Environment

All training, optimization, and evaluation pipelines were executed on a standardized, controlled hardware tier to ensure identical conditions for all models. The computing environment consisted of a single **NVIDIA T4 GPU** equipped with 16GB of High Bandwidth VRAM, provisioned via the Google Colab cloud execution platform. The core runtime environment was driven by Python 3.12, PyTorch 2.1+, and Hugging Face Transformers 4.37.

3 Methods

The experimental procedures were structured to systematically isolate the effects of sequence length, model depth, and attention mechanisms on classification accuracy and training throughput.

3.1 Problem Definition

Given a labeled text classification dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, where each input text x_i consists of a sequence of tokens $\{t_1, t_2, \dots, t_n\}$ and $y_i \in \mathcal{C} = \{0, 1\}$ represents the binary target sentiment class, the objective is to learn a parameterized classifier $f(x_i) \rightarrow y_i$ that minimizes cross-entropy loss.

In the proposed TransMamba-Long framework, the input token sequence x_i is first transformed into a sequence of contextual hidden states via the encoder:

$$H_{enc} = \text{Encoder}_{\text{Longformer}}(x_i) \in \mathbb{R}^{n \times d} \quad (1)$$

where $n = 1024$ represents the maximum sequence length and $d = 768$ represents the hidden state dimension. This hidden space is subsequently channeled into the Mamba decoder block to capture sequential dependencies across the long context horizon:

$$H_{mamba} = \text{Decoder}_{\text{Mamba}}(H_{enc}) \in \mathbb{R}^{n \times d} \quad (2)$$

The representations are then combined through a dedicated cross-attention fusion block, aggregated via mean pooling, and mapped to the label space using a multi-layer classifier head:

$$\hat{y} = \text{Softmax}(\mathbf{W}_c \cdot \text{MeanPool}(\text{Fusion}(H_{enc}, H_{mamba})) + \mathbf{b}_c) \quad (3)$$

3.2 Theoretical and Complexity Analysis

The structural motivation for combining a sliding-window Transformer encoder with a Mamba decoder is rooted in asymptotic computational complexity. Traditional full self-attention architectures require computing a complete pairwise attention matrix, resulting in a time and memory footprint of:

$$\text{Complexity}_{\text{Transformer}} = \mathcal{O}(n^2 \cdot d) \quad (4)$$

For a text sequence scaled to $n = 1024$, this necessitates $1024^2 \approx 1.04 \times 10^6$ basic operations per attention head per layer, causing severe hardware constraints and long training delays.

Conversely, the Mamba layers operate via a parameterized selective state space formulation. By utilizing a recurrent state update hidden mechanism, it updates representations sequentially, restricting the complexity to:

$$\text{Complexity}_{\text{Mamba}} = \mathcal{O}(n \cdot d) \quad (5)$$

By keeping the heavy self-attention restricted to localized windows within the encoder and delegating global long-range tracking entirely to the Mamba decoder, TransMamba-Long successfully maps long documents while maintaining practical and theoretical linearity ($\mathcal{O}(n)$).

3.3 The Selective Scan Mechanism

The core sequence modeling within the Mamba decoder layers relies on input-dependent discretization. Let $x_t \in \mathbb{R}^d$ be the input at token step t . The continuous state space parameters (A, B, C) are mapped to their discrete counterparts $(\bar{A}, \bar{B})$ using an input-dependent step size Δ_t :

$$\bar{A}_t = \exp(\Delta_t A), \quad \bar{B}_t = (\Delta_t A)^{-1}(\exp(\Delta_t A) - I) \cdot B_t \quad (6)$$

The recurrent tracking state $h_t \in \mathbb{R}^d$ and output sequence $y_t \in \mathbb{R}^d$ are then computed via:

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t + D x_t \quad (7)$$

where A is a fixed matrix, while B_t, C_t , and Δ_t are learned linear projections of the input x_t , allowing the network to selectively remember or filter out tokens based on contextual relevance.

3.4 Feature Fusion Pipeline and LoRA Setup

To combine the complementary strengths of the encoder and decoder hidden layers, a multi-head cross-attention fusion module was implemented. The encoder outputs are processed to form the Key (K) and Value (V) matrices, while the Mamba decoder outputs serve as the Query (Q) matrix:

$$F = \text{MultiHeadAttention}(Q = H_{mamba}, K = H_{enc}, V = H_{enc}) \quad (8)$$

The fused representation is integrated with a residual connection and normalized:

$$O = \text{LayerNorm}(H_{mamba} + F) \quad (9)$$

To ensure maximum parameter efficiency and protect the pretraining weights from catastrophic forgetting, **Low-Rank Adaptation (LoRA)** was injected directly into the cross-attention projection layers. The update was restricted to the query and value weights using low-rank decomposition matrices **B** and **A**:

$$\mathbf{W} = \mathbf{W}_0 + \Delta\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r}(\mathbf{B} \cdot \mathbf{A}) \quad (10)$$

where the rank was fixed at $r = 8$ and the scaling hyperparameter was set to $\alpha = 16$. This optimization constrained the total number of trainable parameter weights within the fusion module to approximately 295,000, freezing the remaining base networks.

3.5 Training Configurations and Baselines

To establish an uncompromised baseline comparison, three alternative Transformer models were trained under identical conditions:

1. **DistilBERT (Standard 512)**: Represents the standard, highly efficient encoder-only baseline restricted to its default 512-token context window.
2. **DistilBERT (Overclocked 1024)**: The position embeddings were manually expanded from 512 to 1024 in the PyTorch buffer, and static trackers were reregistered to force the model to ingest the expanded text block.
3. **BART-base (Native 1024)**: A standard sequence-to-sequence Transformer model trained from scratch by Facebook to handle a native 1024-token context window using full quadratic self-attention.

All models—including our proposed 4-layer and 8-layer TransMamba-Long variants—were trained for exactly **1 epoch** to evaluate sample efficiency. The optimization was conducted using the AdamW framework with a strict learning rate of 5×10^{-5} and a batch size of 2 to match hardware limitations and ensure a fair baseline benchmark comparison.

4 Results

The goal of this section is to provide an ordered, objective summary of the quantitative findings recorded during the empirical trials, completely isolated from interpretive analysis. The experimental metrics mapping classification capability against physical training time are detailed in Table 1.

Table 1: Quantitative Performance and Training Latency Comparison on the IMDB Benchmark (Controlled Hyperparameters, 1 Epoch)

Model Architecture	Max Context	Complexity Boundary	Training Time	Accuracy (%)
DistilBERT (Standard)	512	$\mathcal{O}(n^2)$ Quadratic	~21.5 minutes	92.46%
DistilBERT (Overclocked)	1024	$\mathcal{O}(n^2)$ Quadratic	49.54 minutes	91.36%
BART-base (Native)	1024	$\mathcal{O}(n^2)$ Quadratic	48.28 minutes	89.07%
TransMamba (4-Layer)	1024	$\mathcal{O}(n)$ Linear Scan	~18.5 minutes	94.33%
TransMamba (8-Layer)	1024	$\mathcal{O}(n)$ Linear Scan	~35.0 minutes	94.58%

The empirical measurements established that the **4-layer TransMamba-Long** achieved a classification accuracy of **94.33%**, completing its training loop in **18.5 minutes**. The

deeper **8-layer TransMamba-Long** configuration recorded a peak accuracy of **94.58%**, requiring a training runtime of **35.0 minutes** to complete the identical epoch.

Among the baseline architectures, the standard 512-token DistilBERT registered an accuracy score of 92.46% with an execution duration of 21.5 minutes. When forced to process the full 1024 context width, the Overclocked DistilBERT variant registered a degraded accuracy score of 91.36%, while its training clock extended to 49.54 minutes. Similarly, the native 1024-token BART-base architecture recorded an overall accuracy score of 89.07%, consuming 48.28 minutes of training time on the single GPU cluster.

5 Discussion

The primary objective of this section is to interpret the findings and analyze how the combination of State Space tracking and localized Transformer encodings overcomes the historical constraints of sequence classification.

5.1 Deconstructing the Quadratic Attention Penalty

The empirical runtime data laved a clear indictment against traditional global self-attention mechanisms when dealing with long-form context. Doubling the text sequence length from 512 to 1024 tokens caused the training latency of both DistilBERT and BART-base to scale by a factor of **2.3x**, jumping from 21.5 minutes to nearly 50 minutes per epoch. This surge represents a concrete demonstration of the quadratic $\mathcal{O}(n^2)$ bottleneck in practice.

Conversely, by replacing the global sequence attention matrices with recurrent linear selective scanning, the 4-layer TransMamba-Long executed the entire 1024-token context in merely **18.5 minutes**. This indicates a substantial efficiency victory: our proposed long-context architecture ran faster than even the standard 512-token baseline while actively processing twice the context volume.

5.2 Semantic Integrity and Sample Convergence Efficiency

A significant architectural phenomenon was highlighted by the accuracy scores. Manually expanding DistilBERT’s token parameters to 1024 resulted in a severe performance drop, falling from 92.46% down to 91.36%. This structural degradation confirms that forcing an encoder to handle token spaces beyond its pretraining limits introduces uncalibrated spatial noise, which confuses the spatial mapping layers.

Furthermore, while the native 1024 BART-base avoids this noise, it exhibited poor sample efficiency, converging to a subpar 89.07% accuracy within the 1-epoch training constraint. Because large native Transformers contain dense, heavy parameter fields, they require extensive training cycles and massive data iterations to achieve convergence.

In contrast, TransMamba-Long demonstrated exceptional sample efficiency, immediately logging 94.33% accuracy. The underlying integration allows the Longformer encoder to extract clean localized semantics, while the Mamba decoder tracks global long-range text changes without losing focus, establishing superior classification performance right from the initial training loop.

5.3 The 4-Layer Breakthrough: Depth Ablation Analysis

The depth ablation study comparing the 4-layer and 8-layer configurations provided an essential insight for efficient model deployment. While doubling the Mamba decoder layers to 8 established the maximum theoretical accuracy ceiling at 94.58%, this marginal gain of

+0.25% came at a severe computational penalty, causing the training duration to jump from 18.5 minutes to 35.0 minutes (an **89% increase in latency**).

This statistical breakdown demonstrates that the 4-layer variant represents an optimal structural sweet spot. It successfully captures long-sequence dependencies without incurring unnecessary layer duplication, proving that highly optimized structural design is far more valuable than sheer model depth in resource-constrained hybrid systems.

5.4 Task Dependency and Model Limitations

Despite its clear strengths, the study identified specific boundary conditions for the hybrid model. Reviewing past comparative evaluations conducted on short-text frameworks (such as the GLUE SST-2 task, where sequences average fewer than 128 tokens), the hybrid architecture did not consistently outperform standard encoder baselines.

This behavior aligns perfectly with our complexity analysis: on short sequences, the quadratic $\mathcal{O}(n^2)$ attention overhead is not fully activated, meaning the linear advantages of the Mamba decoder become negligible. Therefore, the true value of the Mamba framework only materializes when sequence lengths cross the 512-token threshold.

Additionally, the current framework exhibits a technical limitation: the standard PyTorch SSM library operates via sequential memory allocation, which does not fully leverage hardware-optimized CUDA kernel acceleration. Resolving this software barrier could yield even faster processing speeds in future implementations.

6 Conclusions

This research successfully validated that TransMamba-Long is a highly parameter-efficient framework for long-sequence text classification. By coupling a localized sliding-window Transformer encoder with a lightweight 4-layer Mamba decoder optimized via LoRA, the architecture bypassed the quadratic computational bottleneck of traditional attention mechanisms. The model established a superior scalable framework, securing a high-tier 94.33% accuracy while outperforming native and overclocked baselines in raw processing speed.

Future research lines should extend this hybrid methodology into highly specialized long-document fields, such as legal document tracking (LEDGAR) or scientific abstract classification, to further explore the boundary optimization limits of State Space Models.

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